

LAYER 3 SPECIFICATION · TEMPORAL & FREQUENCY ANALYSIS

Layer 3 - Temporal Consistency & Frequency Analysis

Note (Oct 2026): the algorithm is unchanged. `analyze_frames()` was added so the pipeline passes raw frames (no re-encode), and the face tracker is now reset per clip. Measured behaviour and limitations: [ARCHITECTURE.md](#), Sections 4.4 and 7.

Owner: **Vibha** Part of: Injection Attack & Deepfake Detection pipeline (Layer 1 = Aarya's hardware attestation, Layer 2 = Arihant's PRNU sensor noise profiling, Layer 3 = this module, Layer 5 = gated fusion in Ramya's pipeline)

Files

File	Purpose
<code>temporal_consistency_analysis.py</code>	The Layer 3 detector. Run it on a video and get a JSON verdict.
<code>generate_test_videos.py</code>	Builds two tiny synthetic clips (<code>real_sim.mp4</code> , <code>fake_sim.mp4</code>) so you can sanity-check the analyzer without needing real deepfake footage yet.

How it works

- Per frame, locate the face and compute:
 - a 2D FFT **high-frequency energy ratio** of the face crop
 - edge density** (Canny) as a texture-sharpness proxy
 - mean luminance** as a lighting proxy
- Across a rolling window of frames, look for the tell-tale signs of frame-by-frame generative synthesis:
 - Flicker** - how much those three signals jump around frame-to-frame, relative to their own average level
 - Optical-flow incoherence** - real motion is smooth and spatially consistent; face-swap boundaries often aren't
 - Temporal-FFT periodicity** - take the high-frequency-ratio signal *across time* and FFT it again; a strong periodic peak flags flicker a human eye wouldn't notice
- Combine all four into one **anomaly score in [0, 1]** with a human-readable explanation, in a JSON shape meant to be consumed directly by the pipeline/UI and fused with the other layers.

Run it

```
pip install opencv-python numpy scipy --break-system-packages # if not already installed

python temporal_consistency_analysis.py --video clip.mp4
python temporal_consistency_analysis.py --video clip.mp4 --output result.json
```

Sanity-check without real footage

```
python generate_test_videos.py
python temporal_consistency_analysis.py --video real_sim.mp4
python temporal_consistency_analysis.py --video fake_sim.mp4
```

On these synthetic clips the "fake" one (per-frame injected noise + occasional lighting jumps) currently scores noticeably higher than the "real" one — confirming the pipeline is directionally sound.

Output shape (what Ramya's portal should expect)

```
{
  "layer": "temporal_consistency_frequency_analysis",
  "frames_analyzed": 90,
  "frames_with_face": 90,
  "score": 0.43,
  "flagged": false,
  "confidence": 1.0,
  "components": {
    "flicker_rate": 1.0,
    "mean_flow_coherence": 0.65,
    "flow_incoherence": 0.0,
    "temporal_periodicity_anomaly": 0.46,
    "hf_energy_coefficient_of_variation": 0.03
  },
  "explanation": "...
}
```

`score ≥ 0.6` (configurable) sets `flagged = true`. `confidence` reflects how many frames actually had a detected face vs. the target window size.

Important - this is a heuristic prototype, not a calibrated detector

There's no labeled real-vs-deepfake dataset behind this yet. All thresholds live in the `CONFIG` dict at the top of `temporal_consistency_analysis.py` and are documented placeholders. **Before relying on `flagged` for anything real**, run this against a batch of genuine clips and a batch of known deepfakes (FaceForensics++ is a reasonable public dataset to start from) and retune:

- `hf_instability_norm`, `edge_instability_norm`, `luma_instability_norm`
- `flow_coherence_thresh`
- `temporal_fft_peak_thresh`
- `score_flag_thresh`
- the component `weights`

Integration note for Ramya / Layer 5 fusion

This module only needs a video file path in and returns one JSON object - easy to call as a subprocess or import `TemporalConsistencyAnalyzer` directly and call `.analyze(path)` to get a `LayerResult` object. It doesn't talk to Layer 1 (Aarya) or Layer 2 (Arihant) at all — per the gated-fusion design in the write-up, this layer only needs to run if Layers 1-2 didn't already terminate the session.